Persistent organic pollutants in fish oil supplements on the Canadian market: polychlorinated biphenyls and organochlorine insecticides.

Fish and seal oil dietary supplements, marketed to be rich in omega-3 fatty acids, are frequently consumed by Canadians. Samples of these supplements (n = 30) were collected in Vancouver, Canada, between 2005 and 2007. All oil supplements were analyzed for polychlorinated biphenyls (PCBs) and organochlorine insecticides (OCs) and each sample was found to contain detectable residues. The highest SigmaPCB and SigmaDDT (1,1,1-trichloro-di-(4-chlorophenyl)ethane) concentrations (10400 ng/g and 3310 ng/g, respectively) were found in a shark oil sample while lowest levels were found in supplements prepared using mixed fish oils (anchovy, mackerel, and sardine) (0.711 ng SigmaPCB/g and 0.189 ng SigmaDDT/g). Mean SigmaPCB concentrations in oil supplements were 34.5, 24.2, 25.1, 95.3, 12.0, 5260, 321, and 519 ng/g in unidentified fish, mixed fish containing no salmon, mixed fish with salmon, salmon, vegetable with mixed fish, shark, menhaden (n = 1), and seal (n = 1), respectively. Maximum concentrations of the other OCs were generally observed in the seal oil. The hexachlorinated PCB congeners were the dominant contributors to SigmaPCB levels, while SigmaDDT was the greatest contributor to organochlorine levels. Intake estimates were made using maximum dosages on manufacturers' labels and results varied widely due to the large difference in residue concentrations obtained. Average SigmaPCB and SigmaDDT intakes were calculated to be 736 +/- 2840 ng/d and 304 +/- 948 ng/d, respectively.